**1960** Weber initiates construction of bar detectors for gravitational waves. [Ch. 10]

Kruskal shows that, if it is not threaded by any material, a spherical worm-hole will pinch off so quickly that it cannot be traversed. [Ch. 14]

Graves and Brill discover that the Reissner–Nordstrom solution of Einstein's equation describes a spherical, electrically charged black hole and also a wormhole. [Ch. 7] Their work suggests (incorrectly) that it might be possible to travel from the interior of a black hole in our Universe through hyperspace and into some other universe. [Ch. 13]

**1961** Khalatnikov and Lifshitz argue (incorrectly) that Einstein's field equation does not permit the existence of singularities with randomly deformed curvature, and therefore singularities cannot form inside real black holes or in the Universe's big crunch. [Ch. 13]

**1961–1962** Zel'dovich begins research on astrophysics and general relativity, recruits Novikov, and begins to build his research team. [Ch. 6]

**1962** Thorne begins research under Wheeler's guidance and initiates research that will lead to the hoop conjecture. [Ch. 7]

Giacconi and his team discover cosmic X-rays, using a Geiger counter flown above the Earth's atmosphere on an Aerobee rocket. [Ch. 8]

**1963** Kerr discovers his solution of Einstein's field equation. [Ch. 7]

Schmidt, Greenstein, and Sandage discover quasars. [Ch. 9]

**1964** The golden age of theoretical black-hole research begins. [Ch. 7]

Penrose introduces topology as a tool in relativity research, and uses it to prove that singularities must reside inside all black holes. [Ch. 13]